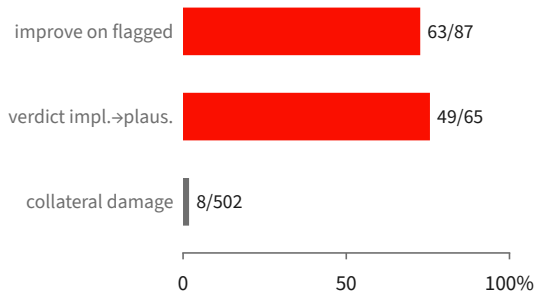


## The loop moves the metric it optimizes, not the independent one

### PhyReAct's own physical metric

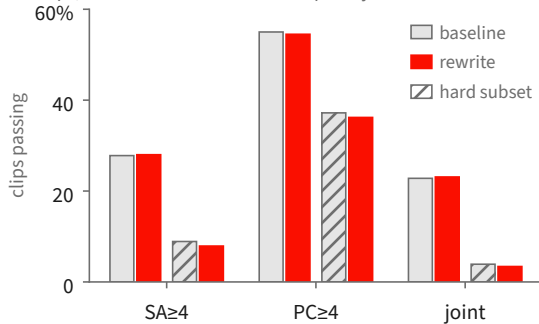
on the clips it flagged — the rewrite's objective



Whole set: mean  $p_{plausible}$  0.797 → 0.862 over all 589 clips (acted on 87/589, mean  $\Delta$  +0.524 on those).

### VideoPhy-2 AutoRater (independent)

same clips, before vs after — not the loop's objective



Left: targeted and low-collateral on its own metric. Right: the independent rater registers no change — either dimension, All or hard.